One FIN Operator, Two Temporal Dynamics

Unitary propagation, reversible diffusion, a finite double-slit experiment, and the operational structure still missing from physics

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Scope. This note studies one precisely defined 12×12 matrix and two functional calculi generated by its weighted graph Laplacian. It does not claim a continuum limit, a physical clock, a microscopic implementation, or a derivation of quantum mechanics from FIN. The double-slit figure is a finite two-path operational analogue on $\mathbb{Z}_{12}$, not a simulation of continuum optical diffraction.

Repository. `https://github.com/hyconiek/Fractal-Nadsoliton-Theory`
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Abstract

Let W be the zero-diagonal cyclic-distance matrix obtained by sampling the strict FIN profile on $\mathbb{Z}_{12}$, let $s = 1.660307278766099$ be its common row sum to fifteen decimal places, and set $A = sI - W$. All six off-diagonal distance weights used at this size are positive. We prove that A is the connected weighted graph Laplacian and that

$$\langle f, Af \rangle = \frac{1}{2} \sum_{x,y} W_{xy} |f_x - f_y|^2 \geq 0.$$

Consequently the same self-adjoint operator generates a unitary group $U_t = e^{-itA}$ and an irreducible, uniform-reversible Markov semigroup $P_t = e^{-tA}$. Their shared spectrum does not make their operational meanings equivalent: unitary position probabilities do not satisfy Chapman–Kolmogorov composition, exhibit quadratic short-time escape and interference, and do not relax to the uniform distribution, whereas P_t has linear short-time jumps and exponentially mixes.

We give a reproducible finite double-slit analogue. A coherent preparation at vertices -2 and $+2$ produces a signed interference term; a which-path record suppresses that term continuously; the Markov model yields a third, distinct law. At the first maximum of total interference, $t_* = 1.5525$, the symmetry detector has unit visibility in the ideal coherent protocol. This is a conditional prediction of an explicit preparation–propagation–measurement model, not a prediction of A alone.

The analysis identifies the smallest standard operational completion capable of carrying the missing roles: a *calibrated process–tester model*, consisting of a causal process tensor (quantum comb), an admissible tester with classical records, and a clock/reference calibration. It recovers state, multitime dynamics, preparation, instruments, environmental memory, apparatus and records up to operational equivalence. It also proves what cannot be recovered from the dimensionless operator: a unique choice of temporal semantics, an absolute time scale, a unique apparatus, or a physical selector. The bridge to experiment is therefore additional operational data, not a theorem of the spectral calculus alone.

Status vocabulary. **[PROVEN]** denotes an analytic statement under the displayed definitions; **[COMPUTED]** denotes a reproducible floating-point result; **[CANDIDATE]** denotes a proposed operational completion grounded in existing theory; **[NOT ESTABLISHED]** marks a deliberately excluded inference.

Contents

1 Executive result

The strongest conclusion that survives the falsification tests is:

> The sampled strict FIN circulant determines a connected weighted graph Laplacian. That Laplacian canonically supports both a reversible heat semigroup and a unitary group with the same Fourier projectors. It does not determine which calculus is physical time. Experimentally meaningful predictions arise only after a preparation, an instrument, a clock calibration, an environment/apparatus model and a record rule are supplied.

The mathematical part is standard finite spectral and graph theory applied exactly to the FIN matrix. The finite double-slit protocol and numerical audit are new computations for this matrix. The proposed operational object is a synthesis of the established quantum-comb, process-tensor, instrument and relational-clock frameworks; it is not claimed as a new foundational theorem.

2 The exact finite operator

2.1 Definition and data convention

Let $V = \mathbb{Z}/12\mathbb{Z}$ and

$$d(x, y) = \min\{|x - y|, 12 - |x - y|\} \in \{0, \ldots, 6\}.$$

Define the sampled profile

$$k_d = \frac{\cos(0.18575d + 0.16250)}{1 + d^{1.8}}, \qquad d = 1, \ldots, 6,$$

and the real symmetric matrix

$$W_{xx} = 0, \qquad W_{xy} = k_{d(x,y)} \quad (x \neq y).$$

Numerically,

$$\begin{aligned} k_1 &= 0.46998567264502017, & k_2 &= 0.19204355169010282, \\ k_3 &= 0.09142861427792497, & k_4 &= 0.04702916874565040, \\ k_5 &= 0.02413122336363006, & k_6 &= 0.011070817321442113. \end{aligned} \tag{1}$$

Every row has the same sum

$$s = 2\sum_{d=1}^{5} k_d + k_6 = 1.660307278766098953\ldots,$$

which rounds to the value in the title data,

$$\boxed{s = 1.660307278766099}, \qquad \boxed{A = sI - W}.$$

Terminological precision. W is the *zero-diagonal all-to-all weighted cyclic-distance matrix* sampled from the strict profile. It is not the adjacency matrix of the nearest-neighbour cycle C_{12}. Nor may directed entries at distances $7, \ldots, 11$ be appended to this definition: cyclic distance folds those separations back to $5, \ldots, 1$. A directed convention would define a different and generally non-self-adjoint operator.

2.2 Dirichlet positivity

Theorem 2.1 (Weighted-Laplacian identity). *For every* $f \in \mathbb{C}^{12}$*,*

$$\langle f, Af\rangle = \frac{1}{2}\sum_{x,y\in V} W_{xy}|f_x - f_y|^2 \geq 0.$$

Moreover, $\ker A = \operatorname{span}\{\mathbf{1}\}$*, so zero is a simple eigenvalue.*

Proof. Symmetry and the common row sum give

$$\begin{aligned}\frac{1}{2}\sum_{x,y} W_{xy}|f_x - f_y|^2 &= s\sum_{x}|f_x|^2 - \sum_{x,y} W_{xy}\overline{f}_x f_y \\ &= \langle f, (sI - W)f\rangle.\end{aligned}$$

Every off-diagonal entry is strictly positive. Equality therefore forces $f_x = f_y$ for all x, y, and constants plainly lie in the kernel. □

Thus A is positive *semidefinite*, not positive definite. The raw matrix W is not positive semidefinite: its smallest eigenvalue is approximately -0.681874762380202.

Proposition 2.2 (Uniqueness of the scalar conservative shift). *For a constant c, the matrix $cI - W$ annihilates* $\mathbf{1}$ *if and only if $c = s$. Hence $sI - W$ is the unique scalar shift of this regular W that is conservative. For a nonregular weighted graph the corresponding construction is $D - W$, with a degree matrix D, not a scalar multiple of I.*

3 One spectrum, two temporal calculi

The Fourier vectors

$$\varphi_m(x) = 12^{-1/2}e^{2\pi i m x/12}, \qquad m = 0, \ldots, 11,$$

diagonalize both W and A. Their eigenvalues are

$$\lambda_m(W) = 2\sum_{d=1}^{5} k_d \cos\left(\frac{2\pi m d}{12}\right) + k_6(-1)^m, \qquad \alpha_m(A) = s - \lambda_m(W).$$

Table 1: Exact Fourier-mode organization and audited numerical spectrum.

Fourier modes	$\lambda_m(W)$	$\alpha_m(A)$
0	1.660307278766099	0
±1	0.906186124559019	0.754121154207080
±2	0.083257764338489	1.577049514427610
±3	−0.301099583210347	1.961406861976446
±4	−0.539261570567111	2.199568849333210
±5	−0.638298993312998	2.298606272079097
6	−0.681874762380202	2.342182041146301

The spectral gap is

$$\gamma = 0.754121154207080.$$

The degeneracy $m \leftrightarrow -m$ is also a direct obstruction to extracting an orientation from this radial operator.

3.1 The reversible Markov semigroup

Theorem 3.1 (Diffusive calculus). *For $t \geq 0$,*

$$P_t = e^{-tA} = e^{-st}e^{tW}$$

is entrywise positive for $t > 0$, symmetric, doubly stochastic and an exact semigroup. Its unique stationary law is $\pi_x = 1/12$, detailed balance holds, and with $\Pi = \mathbf{1}\mathbf{1}^*/12$,

$$\|P_t - \Pi\|_{2\to 2} = e^{-\gamma t}.$$

Proof. Because W is entrywise nonnegative,

$$P_t = e^{-st}\sum_{n\geq 0}\frac{t^n W^n}{n!}$$

is nonnegative. The $n = 0$ term is positive on the diagonal and the $n = 1$ term is positive off the diagonal. The identity $A\mathbf{1} = 0$ gives row stochasticity; symmetry gives column stochasticity and detailed balance. The spectral formula proves the semigroup law, uniqueness of equilibrium and the displayed norm identity. □

The Markov generator is $Q = -A = W - sI$: the rate from y to $x \neq y$ is W_{xy} and the total exit rate is s.

3.2 The unitary group

Theorem 3.2 (Wave calculus). *For $t \in \mathbb{R}$,*

$$U_t = e^{-itA}$$

is a unitary group and $U_t\varphi_m = e^{-it\alpha_m}\varphi_m$. Moreover

$$U_t = e^{-ist}e^{itW}.$$

Thus the scalar shift is a global phase in closed unitary position probabilities, while it is essential for mass conservation in the heat calculus.

The common holomorphic family $F(z) = e^{-zA}$ contains both rays,

$$U_t = F(it), \qquad P_t = F(t),$$

but complex analysis does not select which ray is physical time. Calling this a “Wick rotation” would add physical structure not proved here.

3.3 Their probability laws are inequivalent

Define the Born-position matrix $B_t(x, y) = |U_t(x, y)|^2$. Each fixed B_t is doubly stochastic, but it is not a Markov semigroup.

Proposition 3.3 (Quadratic escape obstructs Chapman–Kolmogorov). *For $x \neq y$,*

$$P_t(x, y) = tW_{xy} + O(t^2), \qquad B_t(x, y) = t^2 W_{xy}^2 + O(t^4).$$

Consequently $B_0' = 0$. If $B_{t+u} = B_t B_u$ held, finite-dimensional semigroup theory would give $B_t = e^{tB_0'} = I$, contradicting $W_{xy} > 0$.

The unitary survival law is

$$B_t(0,0) = 1 - vt^2 + O(t^4), \qquad v = \sum_{x\neq 0} W_{x0}^2 = 0.537963579796275,$$

whereas

$$P_t(0,0) = 1 - st + \frac{1}{2}(s^2 + v)t^2 + O(t^3).$$

The zero linear term is the finite quantum-Zeno signature; the linear Markov loss is a jump rate.

Proposition 3.4 (No nontrivial evolution is both unitary and stochastic)**.** *A continuous path T_t with $T_0 = I$ that is unitary, entrywise nonnegative and stochastic for every t is identically I.*

Proof. A nonnegative unitary matrix is a permutation matrix: orthogonality forces distinct columns to have disjoint supports, leaving one unit entry in every row and column. Permutation matrices form a discrete finite set, so a continuous path starting at I cannot leave I. □

This removes the apparent "observer paradox." The same static A supports two different models. A conscious observer does not transform one law into the other. What changes the predictions is the operational model: state space, preparation, intervening instrument, environment and record rule.

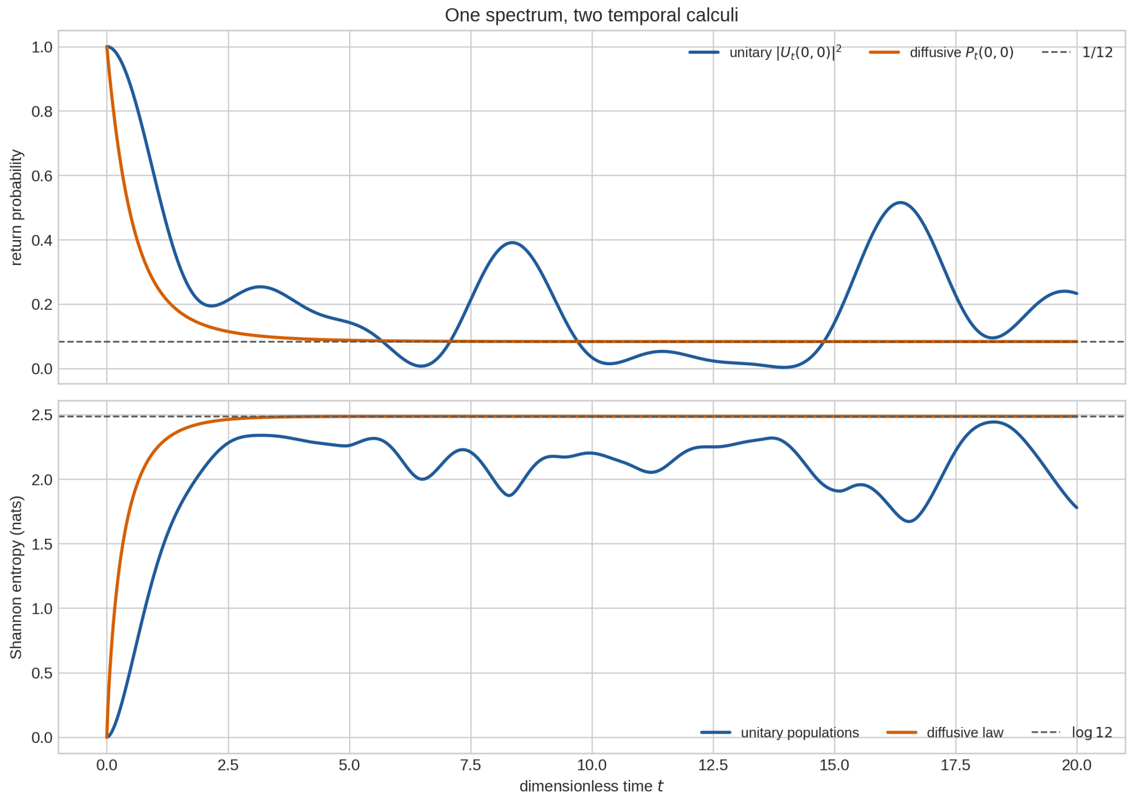

Figure 1: **One spectrum, two temporal calculi.** Starting from a localized vertex, the Markov return probability converges to $1/12$ and its Shannon entropy to $\log 12$. Unitary population entropy is nonmonotone because phase coherence is retained. Time is dimensionless; the plot does not supply a physical clock.

4 A finite double-slit operational experiment

4.1 Protocol

This section implements the requested double-slit simulation in the smallest honest form supported by the matrix. It is a two-path interferometer with a twelve-outcome detector, not a finite-difference approximation to the free Schrödinger equation in Euclidean space.

Choose slit vertices

$$a = -2 \equiv 10 \pmod{12}, \qquad b = +2.$$

A phase-controlled preparation with environmental record states is

$$|\Psi_\phi\rangle = \frac{|a\rangle|E_a\rangle + e^{i\phi}|b\rangle|E_b\rangle}{\sqrt{2}}, \qquad \eta = \langle E_b|E_a\rangle.$$

For the plotted cases $\eta \in [0,1]$ is real. The twelve detector effects are $D_x = |x\rangle\langle x|$. After unitary propagation, write

$$u_a(x,t) = \langle x|U_t|a\rangle, \qquad u_b(x,t) = \langle x|U_t|b\rangle.$$

Tracing out the path record gives

$$p_{\phi,\eta}(x,t) = \frac{|u_a|^2 + |u_b|^2}{2} + \Re\Big(\eta e^{-i\phi} u_a \overline{u_b}\Big). \tag{2}$$

The three compared laws are:

(i) coherent interference: $\eta = 1$ and $\phi = 0$;

(ii) an incoherent quantum mixture / perfectly distinguishing which-path record: $\eta = 0$;

(iii) classical diffusion from $q_0 = (\delta_a + \delta_b)/2$, namely $q_t = P_t q_0$.

Equation (**??**) is the operational bridge in miniature. A supplies U_t, but the phase ϕ, the coherence η, the slit preparation and the detector POVM are independent experimental data.

4.2 Computed result

Define the total interference magnitude

$$\mathcal{I}(t) = \frac{1}{2}\sum_x |p_{0,1}(x,t) - p_{0,0}(x,t)|.$$

The first local maximum on $0 < t \le 5$ occurs at

$$\boxed{t_* = 1.5525}, \qquad \boxed{\mathcal{I}(t_*) = 0.1232806696}.$$

These are grid-resolved floating-point values from the supplied reproducibility script. All three displayed probability vectors sum to 1 within 2×10^{-15}.

At the symmetry detector $x = 0$, reflection invariance gives $u_a = u_b$. Hence

$$p_{\phi,\eta}(0,t) = m_0(t)\,[1 + \eta\cos\phi], \qquad m_0(t) = |u_a(0,t)|^2.$$

At t_*,

$$m_0 = 0.07857926205, qquad p_{\max} = 0.15715852410, qquad p_{\min} \approx 0.$$

The ideal visibility $(p_{\max} - p_{\min})/(p_{\max} + p_{\min})$ is therefore 1. More generally the visibility at that detector is $|\eta|$: an orthogonal apparatus/environment record destroys the fringe, while erasing which-path distinguishability restores it.

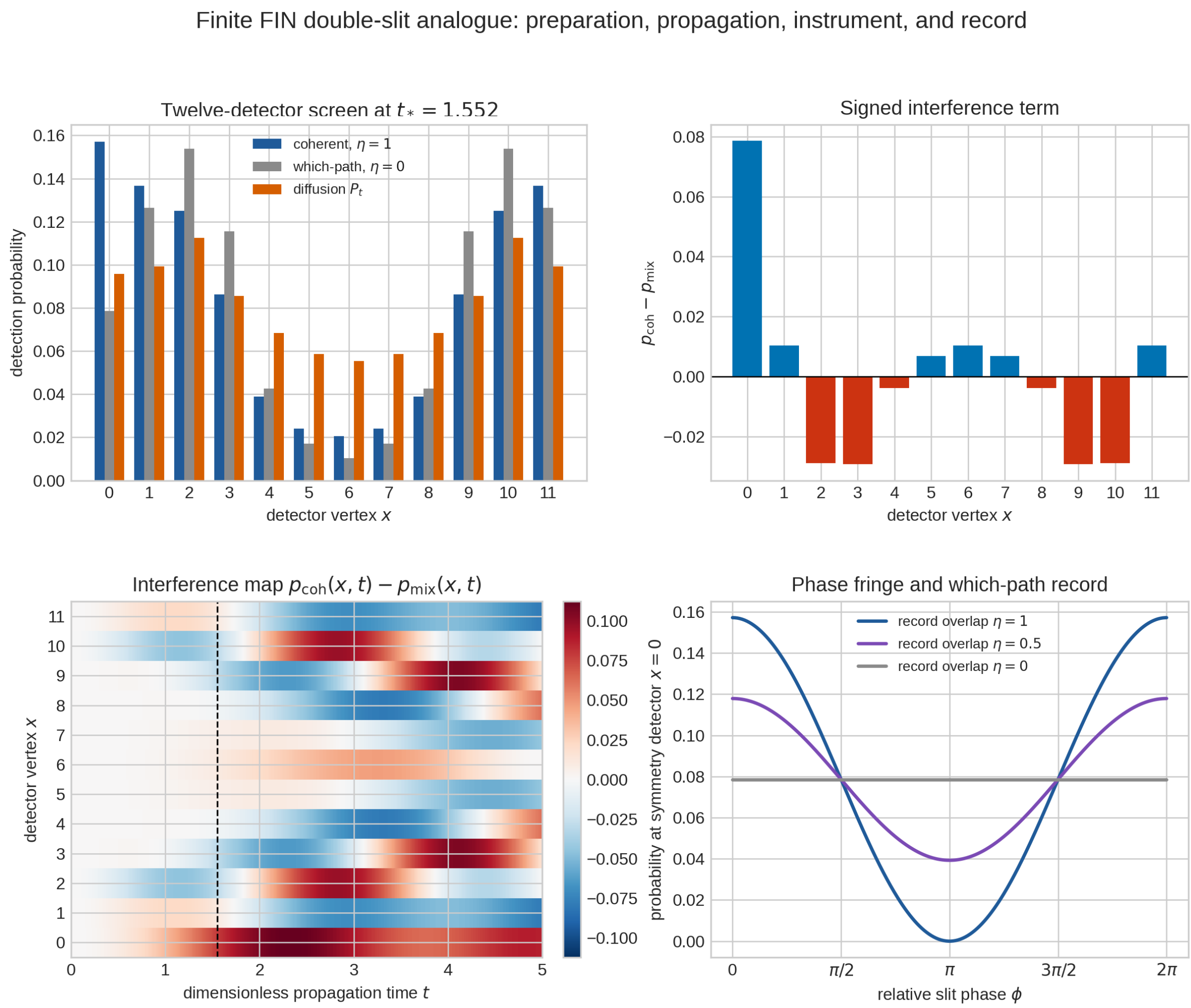

Figure 2: **Finite double-slit analogue generated by** A**.** Top left: the coherent two-slit preparation, the dephased/which-path mixture and classical diffusion at the first interference maximum. Top right: the signed cross term, whose total sum is zero but whose detector-resolved pattern is nonzero. Bottom left: its evolution over all twelve detector outcomes. Bottom right: phase scanning at the symmetry detector; reducing the environment-record overlap η removes the fringe. This figure simulates an explicitly specified operational protocol on $\mathbb{Z}_{12}$, not continuum diffraction.

4.3 Why an intermediate observation changes composition

Without an intermediate measurement, the amplitude is propagated by $U_{t+u} = U_t U_u$ before squaring. With a projective position measurement at time u, the phases are discarded and the observed population law composes as $B_t B_u$. The numerical Frobenius residual

$$\|B_{2t} - B_t^2\|_F$$

is 0.04259 at $t = 0.1$, 0.72319 at $t = 0.5$, and 0.70674 at $t = 1$. By contrast $\|P_{2t} - P_t^2\|_F$ is analytically zero and appears near 10^{-15} numerically. The distinction is an instrument effect, exactly the ingredient absent from a bare operator-only description.

5 Further numerical falsification tests

Table 2: One-site spreading from δ_0. Entropies use natural logarithms and are in nats.

t	$\lvert U_t(0,0)\rvert^2$	$H(\lvert U_t\delta_0\rvert^2)$	$P_t(0,0)$	$H(P_t\delta_0)$
0.1	0.994634304	0.040332484	0.849358649	0.716152665
1	0.584353325	1.293107954	0.262829405	2.225996234
5	0.142549097	2.259728107	0.087250077	2.484375810

5.1 Mixing versus coherent memory

For a localized Markov initial condition, the exact numerical total-variation mixing times are

$$t_{0.1} = 2.4478902, \quad t_{0.05} = 3.3576593, \quad t_{0.01} = 5.4804113, \quad t_{0.001} = 8.5307025.$$

The unitary position law has no pointwise equilibrium. Its infinite-time Cesàro mean is

$$\overline{p} = (11/72, 5/72, 5/72, 5/72, 5/72, 5/72, 11/72, 5/72, 5/72, 5/72, 5/72, 5/72),$$

whose total-variation distance from uniform is $5/36 \approx 0.138889$. The enhancement at vertices 0 and 6 comes from the Fourier degeneracies. Thus even time averaging does not reproduce the Markov equilibrium.

5.2 Why the Perron/Laplacian shift is essential

The raw W has negative eigenvalues. At $t = 1$, e^{-W} has row sum 0.19008056, minimum entry -0.43012412 and operator norm 1.97758175. It is neither stochastic nor contractive. In the unitary model the shift sI changes only global phase; in the diffusive model it is the unique scalar correction that creates conservation. Treating the two shifts as physically interchangeable is therefore false.

5.3 Failure of naive refinement

The fixed analytic profile remains positive at $d = 7$,

$$K(7) = 0.0031528355,$$

but becomes negative at $d = 8$,

$$K(8) = -0.0017958784.$$

A cyclic model first contains distance 8 at $N = 16$. Its off-diagonal Markov generator would then have a negative "rate," and

$$(e^{tQ_N})_{xy} = t(Q_N)_{xy} + O(t^2) < 0$$

for sufficiently small t at that separation. The Markov theorem is therefore certified at $N = 12$ but does not survive naive fixed-parameter refinement. Truncation, rectification, absolute values or parameter running would each be a new axiom.

6 The missing operational object

The user-supplied diagnosis is substantially correct: a physical theory needs a structure that jointly specifies state, dynamics, clock, preparation, instrument, environment, apparatus and measurement record. A bare Hamiltonian or Markov generator specifies none of these as a complete experiment.

6.1 Smallest standard completion

Definition 6.1 (Calibrated process–tester model)**.** A calibrated process–tester model (CPTM) over ordered intervention slots $0, \dots, n$ is the typed object

$$\mathfrak{M} = (\Upsilon_{n:0},\ \mathsf{T},\ \mathsf{C}).$$

Its components are:

(a) a positive, causally normalized process tensor/quantum comb $\Upsilon_{n:0}$, which maps admissible interventions to multitime outcome probabilities;

(b) a tester $\mathsf{T} = \{T_{\mathbf{r}|\mathbf{a}}\}$ made of completely positive instrument elements, including the first preparation slot and explicit classical outcome registers $\mathbf{r}$;

(c) a clock/reference calibration C that associates physical clock readings and a reference frame with the ordered slots and their dimensionless parameters.

The generalized Born rule is the link product, written schematically as

$$p(\mathbf{r}|\mathbf{a}, \mathsf{C}) = \Upsilon_{n:0} * T_{\mathbf{r}|\mathbf{a}},$$

or, under a fixed Choi-transpose convention, as a trace pairing.

This is one object in the sense of a typed operational model, not one unstructured matrix. Process tensors and quantum combs are established mathematics; the adjective "calibrated" records the extra clock/reference datum that these formalisms do not automatically generate.

Theorem 6.2 (Role extraction, conditional on Definition **??**)**.** *A CPTM determines all eight requested operational roles up to their natural operational equivalences:*

Role	*Carrier inside* $\mathfrak{M}$
State	*Initial marginal / zero-slot restriction of* $\Upsilon_{n:0}$
Dynamics	*Conditional multitime maps and memory correlations in* $\Upsilon_{n:0}$
Clock	*Slot order, readings and dimensional calibration in* C
Preparation	*First instrument element in* T
Instrument	*Completely positive outcome-indexed maps in* T
Environment	*Memory system in a minimal dilation of* $\Upsilon_{n:0}$
Apparatus	*A realization/dilation of the tester and reference frame*
Record	*Classical outcome register/pointer algebra indexed by* $\mathbf{r}$

Minimal Stinespring realizations make environment and apparatus unique only up to isometry. A stronger claim of unique microscopic identity is neither operationally necessary nor mathematically available from the observed process alone.

Proof. The entries in the table are restrictions, marginals or dilations of the three typed components. Causal normalization makes probabilities consistent under omission of later instruments; complete positivity makes arbitrary ancillary extensions admissible; the tester supplies the interventions and records; calibration assigns operational time. Standard dilation theorems supply memory/environment and apparatus realizations, unique up to the appropriate isometries. □

6.2 Necessity by deletion

Each component is necessary for the requested coverage.

1. Remove $\Upsilon_{n:0}$: there is no law connecting interventions across time, no initial marginal and no environmental memory class.

2. Remove T: there is no preparation, chosen instrument, apparatus realization or outcome record; A alone gives no experimental probability.

3. Remove C: one retains only an ordered, dimensionless family. Neither a physical duration nor an orientation/reference convention is fixed.

Thus a bare process tensor is still insufficient for the user's full list, and a process plus instruments without calibration remains dimensionless. Conversely, no independent microscopic environment axiom is required for operational predictions because a dilation exists; specifying a particular microscopic bath is additional model detail.

6.3 How A enters, and where it stops

A can seed at least two inequivalent processes:

$$\mathcal{U}_t(\rho) = U_t \rho U_t^*,$$

or the classical Markov process P_t on the commutative position algebra. A Lindblad embedding with jump operators $L_{xy} = |x\rangle\langle y|$ and rates W_{xy} can reproduce the Markov population equation, but that embedding also adds a choice of coherence dynamics. None is selected by A.

The double-slit calculation is a one-step CPTM instance: U_t supplies the process, the two-slit channel supplies preparation, $|x\rangle\langle x|$ supplies the detector, $|E_a\rangle, |E_b\rangle$ supply an environment/apparatus record, and t is the clock parameter. Changing any of these can change observed statistics without changing A.

7 No-go results for deriving physics from A alone

7.1 No absolute scale from dimensionless spectral data

The displayed A is dimensionless. Physical unitary and diffusive evolutions would require, respectively,

$$U_\tau^{\text{phys}} = e^{-i(E_0/\hbar)\tau A}, \qquad P_\tau^{\text{phys}} = e^{-\kappa\tau A},$$

with an energy E_0 or a rate κ.

Proposition 7.1 (Scale obstruction)**.** *No expression built solely from the entries of dimensionless A by algebraic operations, limits and dimensionless spectral functional calculus can select a nonzero dimensional value of time, length, mass, energy or action.*

Proof. All inputs and all operations are invariant under a change of physical unit. A proposed dimensional output would transform nontrivially under the scale group while its input is fixed, contradicting covariance. Equivalently, for every $c > 0$, the pairs (E_0, τ) and $(cE_0, \tau/c)$ give identical dimensionless unitary predictions, and (κ, τ) and $(c\kappa, \tau/c)$ give identical Markov predictions. Nothing in A selects one representative of either orbit. □

Clock calibration may *supply* the missing scale; it cannot be claimed to derive it from A.

7.2 No selector or orientation from the radial operator

W commutes with translations and reflections of $\mathbb{Z}_{12}$. Reflection exchanges modes $+m$ and $-m$ without changing their eigenvalues. Any selector choosing an absolute direction must therefore break a symmetry not broken by A. A chiral preparation or oriented apparatus can provide such a reference operationally, but this is an added resource, not strict-core selector closure.

7.3 Information is not yet thermodynamic entropy

$H(p) = -\sum_x p_x \log p_x$ in Figures **??**–**??** is dimensionless Shannon information in nats. Thermodynamic entropy requires a conversion scale such as k_B and a physical ensemble; work or heat additionally requires temperature and a Hamiltonian/bath model. Landauer-type statements are conditional on those data. The CPTM supplies outcome statistics but does not turn H into energy by itself.

8 Shortest route to an experimental test

A minimal test does not require cosmology or a continuum field theory.

1. Implement twelve controllable modes with a calibrated Hamiltonian $H = E_0 A$ (photonic modes, trapped-ion modes or another programmable interferometer).
2. Prepare $(|10\rangle + e^{i\phi}|2\rangle)/\sqrt{2}$ and scan ϕ.
3. Propagate for $\tau_* = 1.5525\hbar/E_0$ and record the twelve-mode detector distribution.
4. Couple an ancilla to the path so that $\eta = \langle E_b|E_a\rangle$ can be tuned. Verify the predicted linear loss of visibility at detector 0.
5. In a separate classical device, implement rates $Q_{xy} = W_{xy}$ for $x \neq y$ and $Q_{xx} = -s$. Compare its law at $\tau_* = 1.5525/\kappa$ with the Markov curve, without identifying the two physical times unless $E_0/\hbar$ and κ are independently calibrated.
6. Test multitime composition: the Markov transition matrices obey Chapman–Kolmogorov; coherent Born-position matrices do not unless an intermediate measurement is actually made.

This would test the matrix implementation and the operational discrimination theorem. It would not establish that FIN uniquely governs nature: competing Hamiltonians or rate matrices must be included in statistical model comparison.

9 Originality and comparison with existing mathematics

Table 3: What is standard and what is specific to this note.

Topic	Existing framework	Contribution here
Dual functional calculus	Self-adjoint operators generate unitary groups; nonnegative graph Laplacians generate heat semigroups	Exact application, spectrum and boundary audit for the sampled FIN 12×12 matrix
Quantum walks	Continuous-time quantum walks and interference on finite graphs	FIN-weighted two-path predictions and reproducible figures
Classical diffusion	Reversible continuous-time Markov chains on weighted graphs	Exact rates, gap, mixing and refinement obstruction for this matrix
Instruments	Davies–Lewis instruments and POVMs	Identification of the missing preparation/instrument/record data in the FIN claim
Process tensors / combs	General multitime quantum processes with interventions and memory	Calibrated process–tester synthesis covering the eight requested roles
Relational clocks	Operational time from clock correlations/reference systems	Explicit proof that a clock calibration remains extra data

No novelty is claimed for the spectral theorem, graph-Laplacian positivity, quantum interference, Stinespring dilation or process-tensor formalism. The potentially useful contribution is the exact FIN-specific synthesis: it isolates what the operator proves, supplies a falsifiable finite experiment, and locates the irreducible operational and dimensional inputs.

10 Confidence ledger and final answer

Status	Statement
[PROVEN]	$A = sI - W$ is a connected weighted graph Laplacian on the stated 12-vertex cyclic-distance convention.
[PROVEN]	U_t is unitary and P_t is an irreducible uniform-reversible Markov semigroup with shared Fourier projectors.
[PROVEN]	Born-position matrices from U_t are not a nontrivial Markov semigroup; coherent and intermediate-measurement protocols differ.
[PROVEN]	The double-slit formula (**??**) and ideal symmetry-detector visibility follow from the stated protocol.
[COMPUTED]	The spectra, mixing times, Chapman–Kolmogorov residuals, t_* and plotted distributions reproduce to the precision stated in the accompanying JSON file.
[PROVEN]	A dimensionful clock/rate cannot be generated from dimensionless A without a dimensional calibration or equivalent scale resource.
[CANDIDATE]	A calibrated process–tester model is the smallest established-style object found that jointly carries all eight requested operational roles.

Status	Statement
[NOT ESTABLISHED]	A uniquely selects unitary rather than diffusive physical time, a laboratory apparatus, an environment, a Born rule, an orientation, or dimensional constants.
[NOT ESTABLISHED]	The $N = 12$ Markov result has a naive continuum/refinement limit; the fixed profile already violates nonnegative rates at distance 8.

The deepest defensible interpretation is therefore not that an observer chooses between wave and diffusion. It is that A is a common spectral generator whose physical semantics are underdetermined until embedded in a calibrated process–tester model. The missing bridge is operational structure plus dimensional calibration. No theorem of the bare spectral operator can replace those inputs.

Reproducibility

Run

```
python3 fin_dual_dynamics_release.py
```

with Python 3, NumPy and Matplotlib. This regenerates `FIN_Dual_Dynamics_reference_results.json`, `FIN_One_Spectrum_Two_Dynamics.png`, and `FIN_Double_Slit_Operational_Simulation.png`. The JSON records all displayed kernel weights, spectral values, numerical residuals and double-slit diagnostics. Floating-point residuals near 10^{-15} are not physical effects.

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